

TELECOM CUSTOMER CHURN ANALYSIS REPORT

Future Interns – Data Science & Analytics Internship

Task 2: Customer Retention & Churn Analysis

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Tool Used: Tableau, Python (Data Analysis)

Dataset: Telecom Customer Churn Dataset

1. Executive Summary

Customer churn is one of the most critical challenges faced by telecommunication companies because acquiring a new customer is significantly more expensive than retaining an existing one. This project analyzes customer behavior to identify the major factors influencing churn and provides data-driven recommendations to improve customer retention.

Using Tableau, an interactive dashboard was developed to visualize customer behavior, service usage, contract renewal, and churn patterns. The dashboard enables business users to monitor churn trends and make informed strategic decisions.

2. Project Objective

The primary objectives of this project are:

- Analyze customer churn patterns.
 - Identify key drivers influencing customer churn.
 - Measure customer retention.
 - Build an interactive Tableau dashboard.
 - Generate business insights.
 - Recommend strategies to reduce churn.
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3. Dataset Overview

The telecom dataset consists of customer information, service usage, and subscription details.

Key variables include:

- Account Weeks
- Contract Renewal
- Data Plan
- Data Usage
- Customer Service Calls
- Day Calls
- Day Minutes
- Monthly Charge
- Overage Fee
- Roam Minutes
- Churn (0 = Retained, 1 = Churned)

The dataset contains customer-level records used to analyze behavioral patterns associated with churn.

4. Data Preparation

Before analysis, the dataset was cleaned and validated.

The following preprocessing steps were performed:

- Verified data types
- Checked missing values
- Removed duplicate records (if any)
- Validated numerical columns
- Verified target variable (Churn)

The cleaned dataset was then imported into Tableau for visualization.

5. Dashboard Components

The Tableau dashboard includes the following analytical views:

KPI Cards

- Total Customers
- Churn Rate
- Retention Rate
- Average Monthly Charge
- Average Customer Lifetime (Account Weeks)

Visualizations

- Customer Churn Distribution
- Contract Renewal vs Churn
- Customer Service Calls vs Churn
- Monthly Charge Distribution
- Account Weeks Analysis
- Data Plan vs Churn
- Customer Segmentation

Interactive Filters

- Contract Renewal
- Data Plan
- Churn Status

These filters allow users to explore customer behavior dynamically.

6. Business Insights

The dashboard reveals several important insights.

Insight 1

Customers who did not renew their contracts show a noticeably higher likelihood of churning.

Insight 2

An increase in customer service calls is associated with a higher churn rate, indicating that unresolved service issues may contribute to customer dissatisfaction.

Insight 3

Customers with higher monthly charges tend to exhibit increased churn, suggesting pricing sensitivity.

Insight 4

Longer-tenure customers (measured using Account Weeks) generally demonstrate stronger retention.

Insight 5

Customers with active data plans display different usage patterns, providing opportunities for targeted retention campaigns.

Insight 6

Overage fees may negatively influence customer satisfaction and contribute to churn.

7. Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Develop proactive retention programs for customers at high risk of churn.
 - Encourage contract renewals through personalized offers and loyalty incentives.
 - Improve customer service response times and issue resolution.
 - Review pricing strategies for customers with high monthly charges.
 - Reduce unnecessary overage fees through optimized plans.
 - Introduce customized offers based on customer usage behavior.
 - Monitor high-value customers and intervene before potential churn.
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8. Business Impact

Implementing these recommendations can provide several business benefits:

- Reduced customer churn
 - Increased customer loyalty
 - Improved customer satisfaction
 - Higher recurring revenue
 - Better customer lifetime value
 - More efficient marketing campaigns
 - Enhanced decision-making using interactive dashboards
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9. Tools and Technologies

The following tools were used during the project:

- Tableau Desktop
 - Python
 - Pandas
 - NumPy
 - Matplotlib
 - Jupyter Notebook
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10. Conclusion

This project demonstrates how data analytics and business intelligence can support customer retention strategies in the telecommunications industry. The Tableau dashboard provides a clear and interactive view of churn behavior, enabling stakeholders to identify at-risk customers and understand the factors contributing to churn.

The insights generated through this analysis can assist decision-makers in designing targeted retention initiatives, improving customer satisfaction, and increasing long-term profitability. The project also highlights the value of combining exploratory data analysis with interactive visualizations to transform raw data into actionable business intelligence.

Future Scope

Future improvements could include:

- Predictive churn modeling using machine learning.
 - Customer lifetime value prediction.
 - Real-time dashboard updates.
 - Integration with CRM systems.
 - Personalized customer retention recommendations using AI.
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End of Report